PROJECT CHARTER: Project Dalupotha

1. Project Overview

Project Name: Project Dalupotha (Tea Supply Chain & Factory ERP) **Project Description:** A centralized, digital ecosystem to automate and manage the collection of green tea leaves from Small Holders, streamline transport logistics, and provide transparent access to factory services, accounting, and inventory.

2. Project Purpose & Justification

Business Problem: The factory currently relies on fragmented, manual procedures for recording green leaf collections, managing financial advances, and issuing stores inventory. This results in data discrepancies, delayed supplier payments, limited logistical visibility, and poor accessibility to vital agricultural services for Small Holders. **Justification:** By digitizing the end-to-end workflow from field collection to factory processing, the factory will minimize operational bottlenecks, eliminate manual entry errors, optimize transport routes, and significantly improve supplier loyalty through transparent, rapid service delivery.

3. Measurable Objectives & Success Criteria

- **Accuracy:** Achieve 100% automated logging of green leaf weights directly from IoT/Bluetooth-enabled hanging scales, eliminating manual recording errors.
- **Traceability:** Capture verified GPS coordinates for 100% of Small Holder registrations and Transport Agent collection points.
- **Efficiency:** Reduce the processing time for advances, balance payments, and fertilizer requests by integrating these workflows into a centralized portal.
- **Adoption:** Ensure continuous field operation by designing a mobile application with robust offline-first capabilities that successfully syncs data upon network reconnection.

4. High-Level Requirements & System Architecture

Core Functional Requirements:

- **Role-Based Access Control:** Secure portals for Managers (MG), Small Holders, Office Staff (ST), Store Keepers (SK), Factory Staff (FT), Extension Officers (EXT), and Transport Agents (TA).
- **Transport & Field Operations:** A mobile app for Transport Agents featuring GPS tracking, offline data synchronization, and automated weight capturing.

- **Small Holder Portal:** OTP-based login for suppliers to view green leaf supply history, advance histories, balance payments, debts, and to submit requests for fertilizer, leaf bags, and transport.
- **Factory & Office Operations:** Centralized dashboards for managing the brought leaf register, local sale tea, machinery renting, TRI circular distribution, and overall quality deductions.

Architecture Strategy: To ensure scalability and maintainability, the solution should be designed as a modular, plug-and-play ERP system.

- **Integration:** The core operational modules will be structured to interface seamlessly with dedicated accounting systems (such as Ginum) to automate balance and advance calculations, as well as HR solutions (like PirisaHR) for managing factory and extension staff.
- **Infrastructure:** The backend and database architecture will leverage robust cloud environments—such as Supabase to handle the complex offline-sync requirements of the field app and provide real-time updates—deployed across scalable infrastructure like Digital Ocean, OCI, or Azure to guarantee high availability during peak harvesting seasons.

5. Scope Boundaries

In-Scope:

- Development of the Transport Agent mobile application (collection, GPS, IoT scale integration).
- Development of the Management/Office web dashboard (inventory, approvals, reporting).
- Development of the Small Holder service portal (OTP login, history viewing, service requests).
- Supplier registration module with geographic mapping.

Out-of-Scope:

- Internal manufacturing process control (e.g., controlling factory machinery temperatures/drying times).
- Direct employment management of the Small Holders (they remain independent suppliers).

6. Key Deliverables

1. **System Architecture & UI/UX Designs:** Wireframes prioritizing simplified navigation for varying digital literacy levels.
2. **Transport Agent Mobile Application:** Android/iOS app with Bluetooth IoT connectivity and local caching.
3. **Dalupotha Central Web Platform:** The core ERP dashboard for office and management staff.
4. **Hardware Procurement Plan:** Specifications for ruggedized mobile devices, thermal printers, and Bluetooth hanging scales.

7. Project Constraints & Assumptions

- **Constraints:** Poor cellular network coverage in remote tea estates strictly mandates a flawless offline-first data capture architecture. Field hardware must be weather-resistant and capable of full-day battery life.
- **Assumptions:** Small Holders will have access to basic smartphones to receive OTPs and access the web portal. Transport Agents will comply with standard operating procedures for utilizing digital scales and GPS tracking.

8. Key Stakeholders

- **Small Holders:** Primary suppliers and service receivers.
- **Transport Agents (TA):** Responsible for executing field logistics, collecting tea, and logging initial data.
- **Extension Officers (EXT) & Extension Managers (MG):** Oversee supplier relationships, advances, and advisory requests.
- **Store Keepers (SK):** Manage inventory for fertilizer and leaf bags.
- **Office Staff (ST) & Factory Staff (FT):** Process daily operational data, payments, and brought leaf registries.
- **Management (MG):** Monitor overall operations, provide approvals, and utilize strategic dashboards.